# PENGFEI ZHANG

Applying for 2027 New Grads with F1 OPT (Expected Graduation Time: Mar 2027 – Jun 2027)

Email: **pengfz5@uci.edu** ◇ Github ◇ LinkedIn ◇ Personal Website

## EDUCATION

**PhD Student in Computer Science** — Fall 2022 – Present
University of California, Irvine — GPA: 3.93/4.0

**B.Eng in Computer Science** — 2017 – 2021
University of Science and Technology of China — GPA: 3.45/4.3

## RESEARCH INTEREST

Human Understanding, Multimodal Large Language Models, Multimodal Agentic RL

## RESEARCH PROJECTS

### I. VLM-based Multimodal Agents for Human-Centric Image/Video Understanding, Generation, and Editing

*This project focus on the understanding, generation, and editing of human-centric images or videos with complex motions, as essential personalized functions of Multimodal LLMs and Multimodal Agents.*

**Techniques:** VLM-based Multimodal Agents Orchestration, MLLM Post-Training (DPO, GRPO, GDPO)

(e) **Identity Decay in Multi-Turn Human-Centric Image Editing: A Benchmark and a Multimodal Agentic Correction Framework** *Under Review*

- Introduced a benchmark that measures the identity and fidelity decay for multi-turn Human-Centric Image Editing.
- Proposed a correction pipeline in which a multimodal agent orchestrates the identity correction process using pre-trained human segmentation and parsing models (sapiens for body, DML_CSR for face), and deterministic photoshop and image operators (such as tps warp, high-freq preservation transfer, etc), flattening the degradation curve.

(d) **KinMo: Kinematic-aware Human Motion Understanding and Generation** **ICCV 2025**

- Designed a VLM-based dataset enhancement pipeline that extends the existing text-motion alignment dataset by incorporating proposed motion representation as body decomposition, enabling precise motion descriptions.
- Introduced a hierarchical motion-text alignment approach that fuses motion details into text semantics, improving SOTA text-motion understanding and text-based motion generation models with more precise motion control.

(c) **LLM-Based Human-Agent Collaboration and Interaction Systems: A Survey** **ACL 2026**

(b) **Contextual Gesture: Co-Speech Gesture Video Generation through Context-aware Gesture Representation** **ACM MultiMedia (MM) 2025**

- A framework that improves co-speech gesture video generation through chronological speech-gesture alignment that temporally connects two modalities, and a contextualized gesture tokenization that incorporate speech context into motion pattern representation through distillation.

(a) **Handformer2T: A Lightweight Regression-based model for Interacting Hands Pose Estimation from a single RGB Image** **WACV 2024**

- A lightweight model which hand-level tokenization in the transformer based model for interacting hand pose estimation, where only one token was used for each hand.

### II. Speech Language Models and Agents in Low-Resource Settings

*This project focus on the joint understanding of multimedia signals (texts, videos, speeches, audios, etc), especially in low-resource settings.*

**Techniques:** MLLM Post-Training (DPO, GRPO, GDPO), Chain-of-thought (CoT) Distillation

(f) **AP-GRPO: Anchor-Gated Phonetic Alignment with Policy Optimization for Pathological Speech Reconstruction** **EMNLP 2026 (accepted)**

- A GRPO framework with phonetic reward that aligns speech language models (SLMs) through audible-anchor preservation and inter-anchor phonetic compatibility to the original speech signal. AP-GRPO consists of: (i) an anchor-gated reward that matches anchors in clear regions; and (ii) an inter-anchor phonetic alignment reward that evaluates whether recovered contents are phonetically supported by the corresponding corrupted inter-anchor speech span.

(e) **DemMA: Dementia Multi-Turn Dialogue Agent with Expert-Guided Reasoning and Action Simulation.** **ACL 2026**

- An expert-guided multi-turn dialogue agent designed to simulate dementia patients with clinically grounded personas by distilling expert-guided reasoning and action simulation.

(d) **MedSpeak: Knowledge Enhanced ASR Error Correction framework for Spoken Medical Question Answering.** **ICASSP 2026**
· a knowledge-enhanced retrieval augmented generation (RAG) framework through a knowledge graph (KG) capturing semantic medical relationships and phonetic similarities between medical terms.

(c) **AudioLens: Multi-Perspective Speech Clustering with Audio-Language Models** *Under Review*
· Introduce audio multi-perspective clustering, where a model directly partitions speech recordings according to a natural-language perspective while inferring both the number of clusters and their assignments. Construct AudioLens-Bench, a benchmark spanning multiple applications and evaluating cross-perspective generalization, and propose AudioLens-R1, a large audio-language model trained with reasoning distillation and direct preference optimization.

(b) **From Sounds to Scenes: A Benchmark for Evaluating Context-Aware Auditory Scene Understanding in Large Audio Language Models.** *Under Review*
· A benchmark designed to evaluate whether Large Audio-Language Models can jointly interpret speech, events, and scene-level information to perform coherent semantic reasoning.

(a) **Hallucination Rates of Speech Language Models on Pathological Speech Recognition: A Pilot Multi-Cohort Audit.** **BSN 2025**
· An evaluation of popular Speech Language Models (SLMs) on Pathological Speech Recognition at different hallucination types (insertion, fluent completion, entity fabrication, semantic substitution).

## PROJECTS IN PRODUCTION

### I. Tiktok Ads – Chatgpt App

· A conversational advertisement creation app on the ChatGPT App Store, that helps novice advertisers turn a product page URL into a live Tiktok Smart+ video clips.
· This app uses chatgpt as orchestrator and conversational agent, with internal skills which scrape account's product details to generate videos using backend Seedance API.
· Now on beta test for 10k+ advertiser registered in Tiktok Business, Customer Retention Rate $> 55\%$
· **Techniques:** MCP Multi-Layer (Bytedance Infra), Multi-step Conditioned Video Generation, Agent Self Reflection, Agent Orchestration

### II. Claworld

*An Agent-to-Agent Platform for agents to interact with each other, trading skills or carrying on tasks.*

· An online platform with consistent policy for agents to trade their skills, tokens, and efforts.
· **Techniques:** Agent Society

## WORKING EXPERIENCE

**Bytedance** Mar. – Sep. 2026
*Machine Learning Engineer Intern, Agent Monetization Team*
· Multimodal Agents Tool Usage and Social Intelligence of Agents (Publications I.e, Proj I, II).

**Amazon Web Services (AWS AI Lab)** Jun. – Sep. 2025
*Applied Scientist Intern, AI Transcribe Team*
· Speech LLMs recognition, understanding, and reasoning in low-resource languages (Publications II.a, II.f).

**Flawless. AI. Inc** Jun. – Sep. 2024
*Research Science Intern, Intelligent AI Team*
· Multimodal LLMs for Human Motion Understanding and Multimedia Generation/Editing (Publications I.c, I.b).

## AWARDS

Dean's award from UCI 2022 – 2024
National Encouragement Scholarship (top 20%) from USTC 2018 – 2020